

CHE 201 Fall 2019 Quiz 1

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TOTAL POINTS

90 / 100

QUESTION 1

1 Question 1 **40 / 40**

✓ - **0 pts** Correct

- **10 pts** Patm is wrong
- **5 pts** unit conversion
- **15 pts** Pgage is wrong or not calculated
- **15 pts** P absolute is wrong or not calculated
- **40 pts** Not done or incorrect

QUESTION 2

Question 2 60 pts

2.1 a **20 / 30**

- **0 pts** Correct
- ✓ - **5 pts** system not identified
- ✓ - **5 pts** engineering model is missing
- **10 pts** PV diagram is wrong
- **15 pts** Work calculation is wrong
- **5 pts** unit is not converted
- **30 pts** not done

2.2 b **30 / 30**

- ✓ - **0 pts** Correct
- **20 pts** incorrect or no calculation
- **10 pts** PV diagram is wrong
- **0 pts** Click here to replace this description.

NAME: Ishak Almesmari UIN: _____

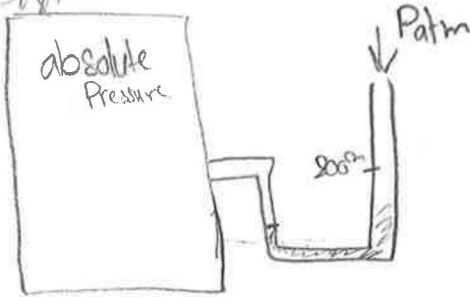
QUIZ 1: 2 problems (10 pts)

Closed book/Closed notes: Show all your work for full credit

1. (6 pts) The average atmospheric pressure on Earth is approximated as a function of altitude by the relation $P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256}$, where P_{atm} is atmospheric pressure in (kPa) and z is altitude in km with $z=0$ at sea level. A monometer is used to measure the pressure of a gas in a tank in Denver ($z=1610\text{m}$). The fluid has a density of 850kg/m^3 and the monometer column height is 200cm . Determine the absolute and gage pressure of gas within the tank.

knd b d cm

$$\frac{\text{kg m}}{\text{s}^2 \text{m}^2} = \frac{\text{kg}}{\text{s}^2 \text{m}}$$



$$\rho = 850 \text{ kg/m}^3$$

$$h = 200 \text{ cm} = \cancel{0.2 \text{ m}} 2 \text{ m}$$

$$P_{\text{atm}} = 101.325 \left(1 - 0.02256 \right)^{5.256} \times 1.610$$

$$1 \text{ Pa} = 1 \text{ N/m}^2 \Rightarrow \text{N} = \text{kg m/s}^2$$

$$P_{\text{atm}} = 83.42 \text{ kPa}$$

$$P_{\text{gas}} = P_{\text{atm}} + \rho g L = 83.42 \text{ kPa} + \left(850 \frac{\text{kg}}{\text{m}^3} \right) \left(9.8 \frac{\text{m}}{\text{s}^2} \right) (2 \text{ m})$$

$$P_{\text{gas}} = 83.42 \text{ kPa} + 16660 \text{ Pa}$$

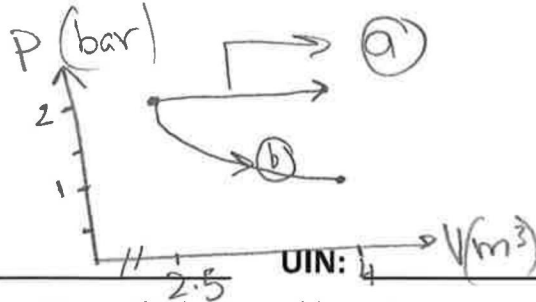
$$1 \text{ Pa} = \frac{\text{kg m}}{\text{s}^2 \text{m}^2}$$

$$= 83.42 + 16.660 \text{ kPa} = 100.08 \text{ kPa}$$

$$P_{\text{gage}} = P_{\text{absolute}} - P_{\text{atm}} = 100.08 - 83.42 \text{ kPa}$$

$$P_{\text{gage}} = 16.66 \text{ kPa}$$

NAME: _____



UIN: _____

2. (4 pts) A gas contained within a piston-cylinder assembly undergoes an expansion process for which the relationship between pressure and volume is given by

$PV^n = \text{constant}$

The initial pressure is 2 bar, $V_1 = 2.5 \text{ m}^3$ to $V_2 = 4 \text{ m}^3$. Determine the work for the process

in kJ, if

a. $n = 0$ and

b. $n = 1.5$

c. Sketch both processes on a P-V diagram

Btq
Pascal

a) $P = \text{constant}$

so $P_2 = 2 \text{ bar}$

$P = \text{constant} = 2 \text{ bar}$

$W = \int_{V_1}^{V_2} P \cdot dV \Rightarrow P = 2 \text{ bar}$

$W = 2 \text{ bar} (V_2 - V_1) = 2 \times 10^5 \frac{\text{N}}{\text{m}^2} (4 - 2.5) \text{ m}^3$

$W = 300,000 \text{ J} = 300 \text{ kJ}$

b) $PV^{1.5} = \text{constant}$

$2(2.5)^{1.5} = P_2(4)^{1.5}$

$P_2 = 0.99 \text{ bar}$

$PV^{1.5} = \text{constant} = 2 \times 10^5 \times (2.5)^{3/2} =$

$W = \int_{2.5}^4 \frac{790569.415}{V^{3/2}} \cdot dV$

constant = $790569.415 \text{ Pa} \cdot (\text{m}^3)^{3/2}$

$W = 790569.415 \left[\frac{2V^{-1/2}}{-1/2} \right]_{2.5}^4 = 790569.415 \left[-4\sqrt{V} \right]_{2.5}^4$

$W = 1790569.415 \text{ J} = 1790.6 \text{ kJ}$

$W = 209430.6 \text{ J} = 209.43 \text{ kJ}$